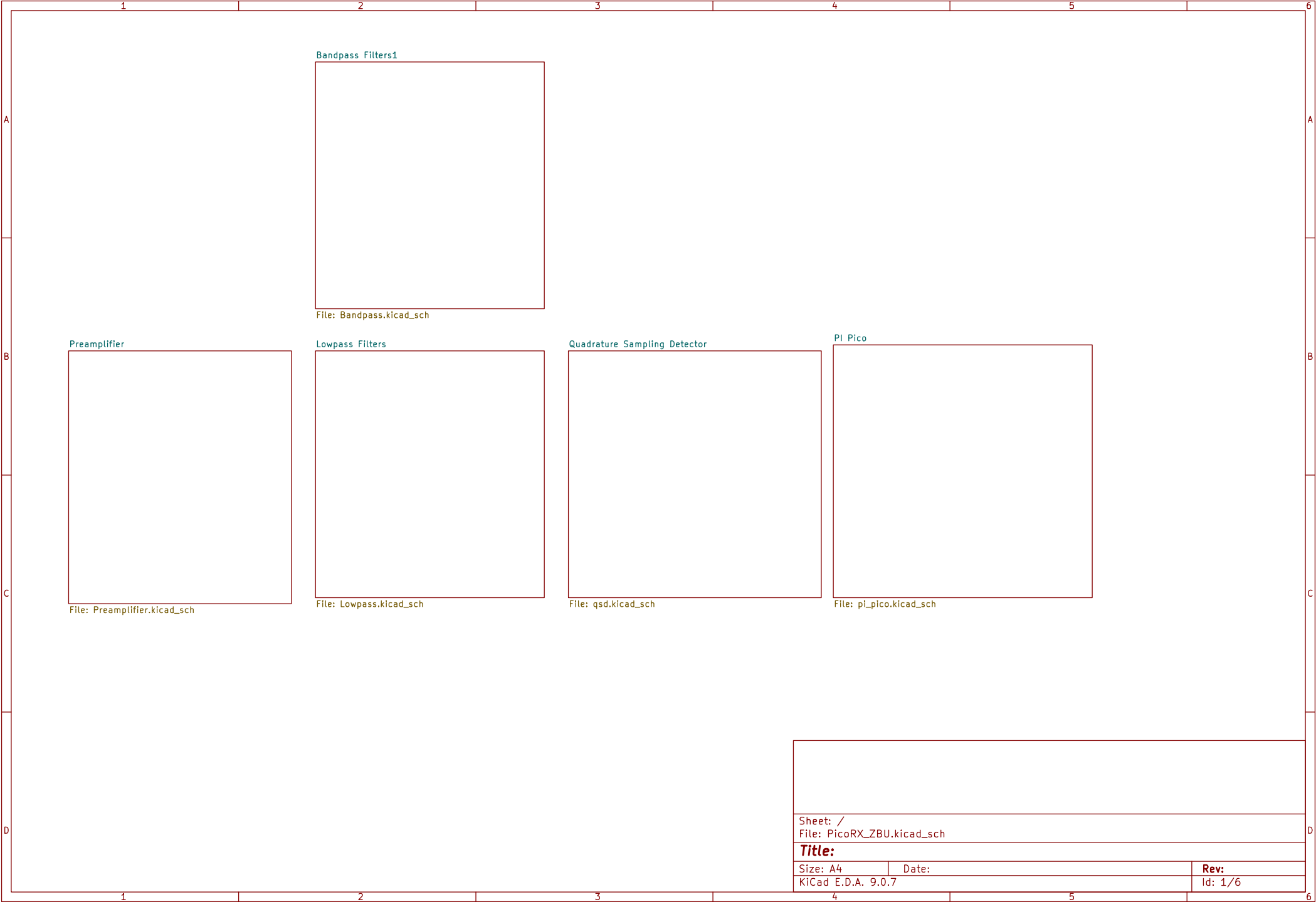


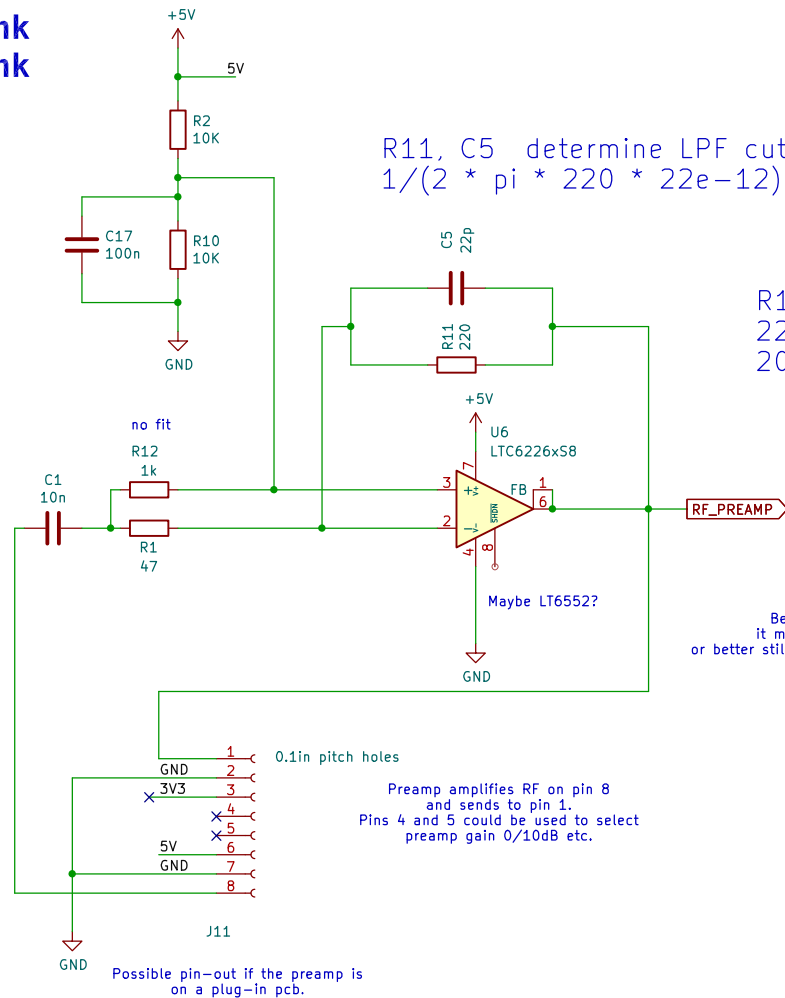
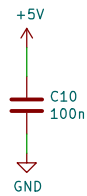


WITH PREAMP

R12 no fit
R1 47
R11 220
C5 22p
U6 fit

WITHOUT PREAMP

1k
0 ohm link
0 ohm link
no fit
no fit



R11, C5 determine LPF cutoff
 $1/(2 * \pi * 220 * 22e-12) \approx 30\text{MHz}$

R1, R11 determine gain
 $220/47 = 5\times$ gain
 $20 * \log_{10}(220/47) = 13\text{dB}$ gain

Because the MUX needs a 50 ohm driver,
it may be worth having a unity gain buffer IC,
or better still, a switchable 0dB/10dB gain 0-30MHz amplifier.

Preamp amplifies RF on pin 8
and sends to pin 1.
Pins 4 and 5 could be used to select
preamp gain 0/10dB etc.

Possible pin-out if the preamp is
on a plug-in pcb.

Sheet: /Preamplifier/
File: Preamplifier.kicad_sch

Title:

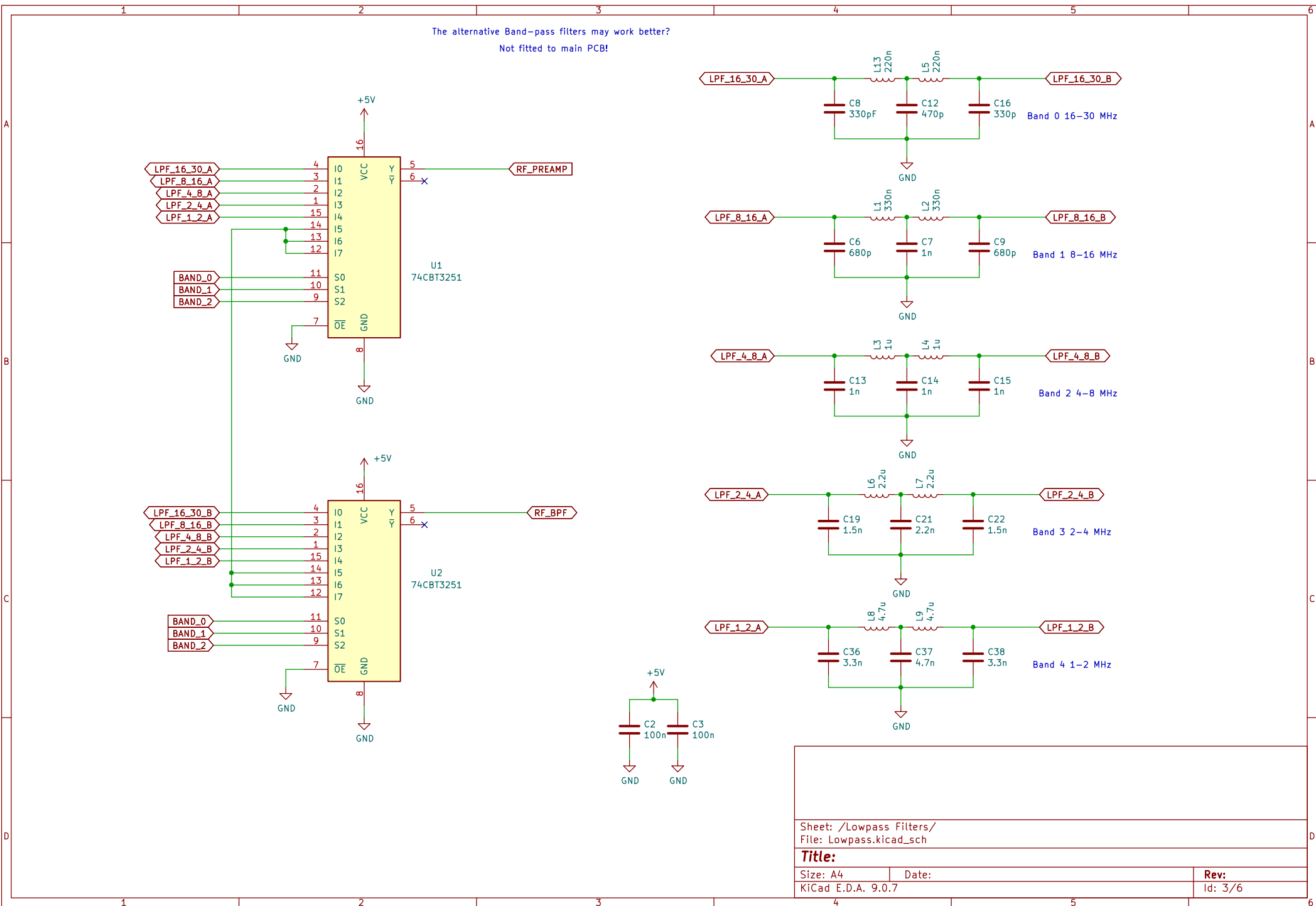
Size: A4

Date:

Rev:

KiCad E.D.A. 9.0.7

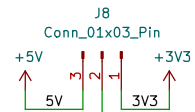
Id: 2/6



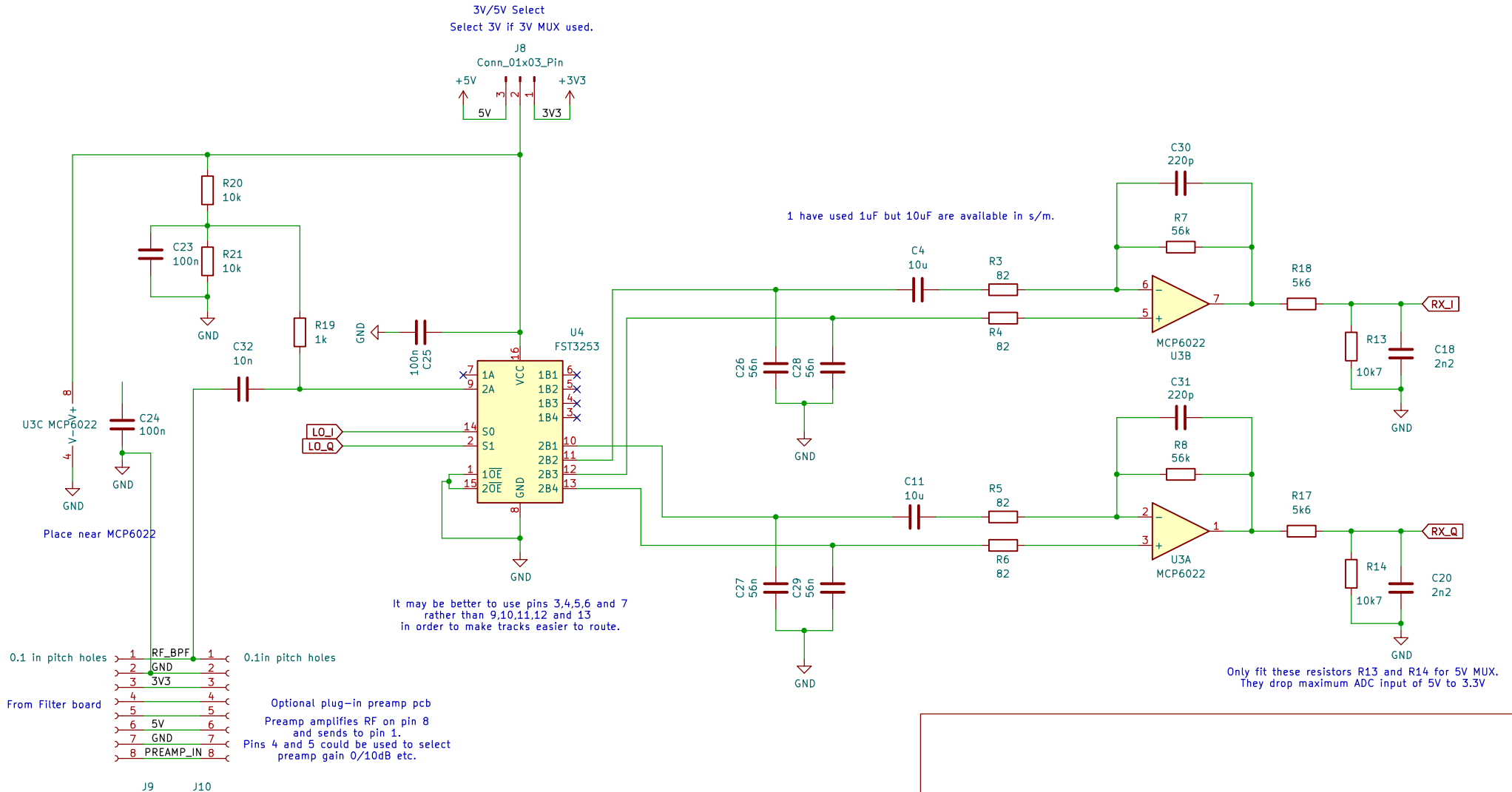
R, C values determine LPF cutoff
 $1/(2 * \pi * 27e-9 * 100) \approx 59\text{kHz}$
 or 56k and 220pF $\approx 12\text{kHz}$

R, values determine gain
 $100\text{k}/100 = 1000\times$
 $20 * \log_{10}(100\text{k}/100) = 60\text{dB}$

3V/5V Select
 Select 3V if 3V MUX used.



1 have used 1uF but 10uF are available in s/m.



RF goes into pin 1 if no preamp.
 Otherwise RF fed into pin 8.

Sheet: /Quadrature Sampling Detector/
 File: qsd.kicad_sch

Title:

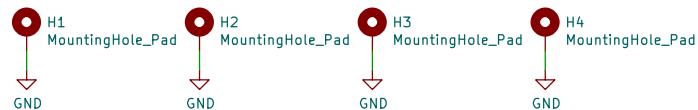
Size: A4

Date:

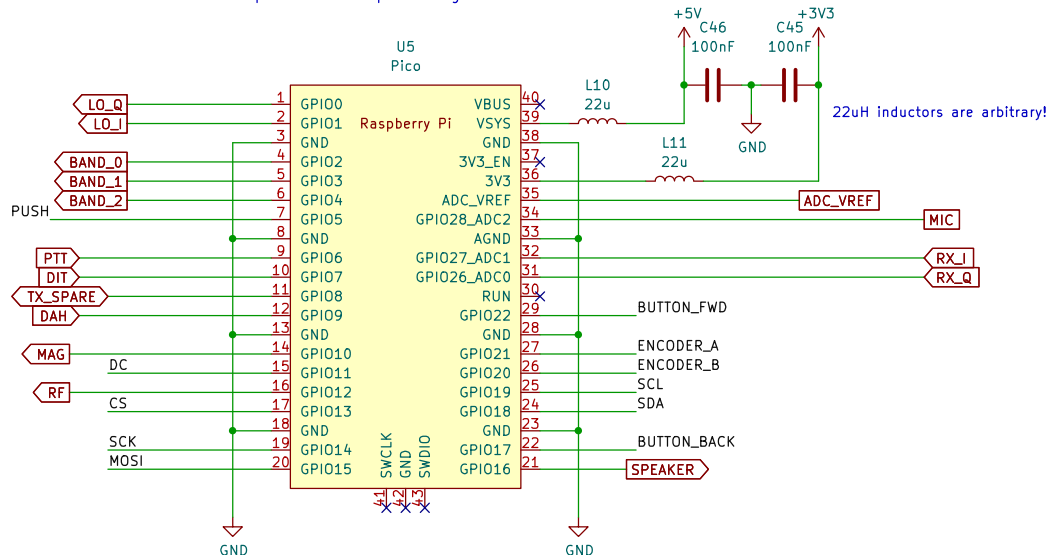
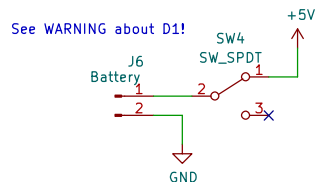
Rev:

KiCad E.D.A. 9.0.7

Id: 4/6



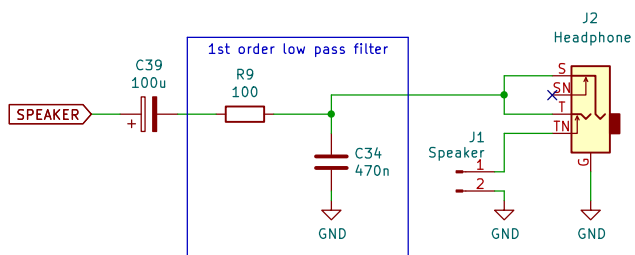
Note: D1 must be removed from PI Pico if 3xAA cells are used to power RX.
Otherwise the USB power will attempt to charge them at an infinite current!



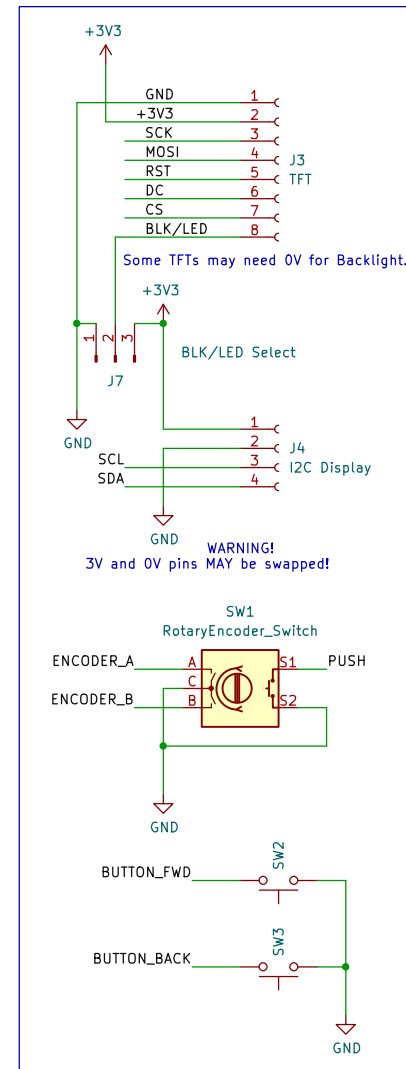
Signals for possible transmit functionality:
MIC, PTT, DIT, DAH, RF and MAG

I have added TX_SPARE for future-proofing!

R9, C34 determine LPF cutoff
 $1/(2 * \pi * 100 * 470e-9) \approx 3\text{kHz}$



User Interface



Sheet: /PI Pico/
File: pi_pico.kicad_sch

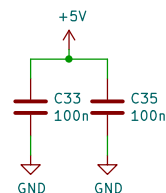
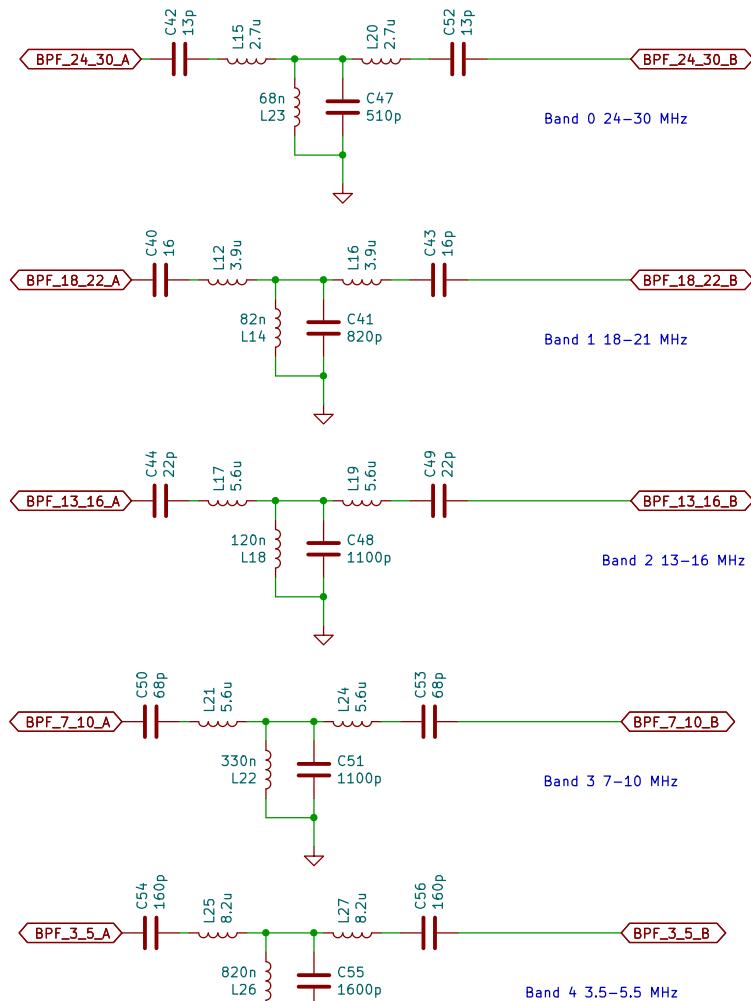
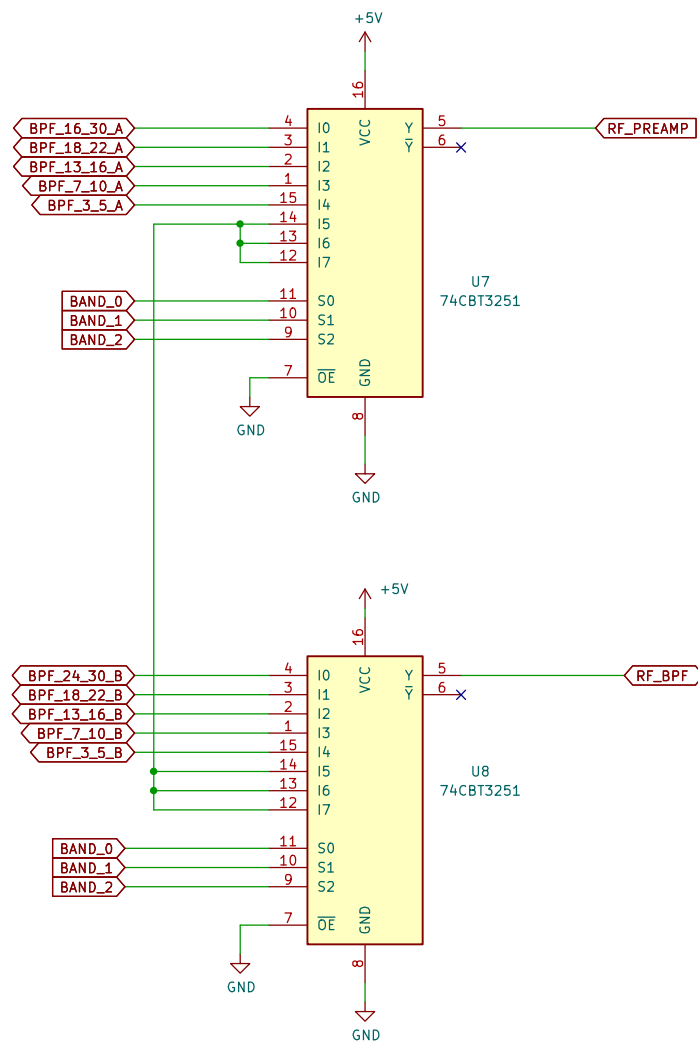
Title:

Size: A4
KiCad E.D.A. 9.0.7

Date:

Rev:
Id: 5/6

Use these instead of Low-pass filters?
 Values need checking with LTSpice.
 Not fitted to main PCB!



Another BPF for 160m ???

<https://markimicrowave.com/technical-resources/tools/lc-filter-design-tool/>

Sheet: /Bandpass Filters1/
 File: Bandpass.kicad_sch

Title:

Size: A4

Date:

Rev:

KiCad E.D.A. 9.0.7

Id: 6/6